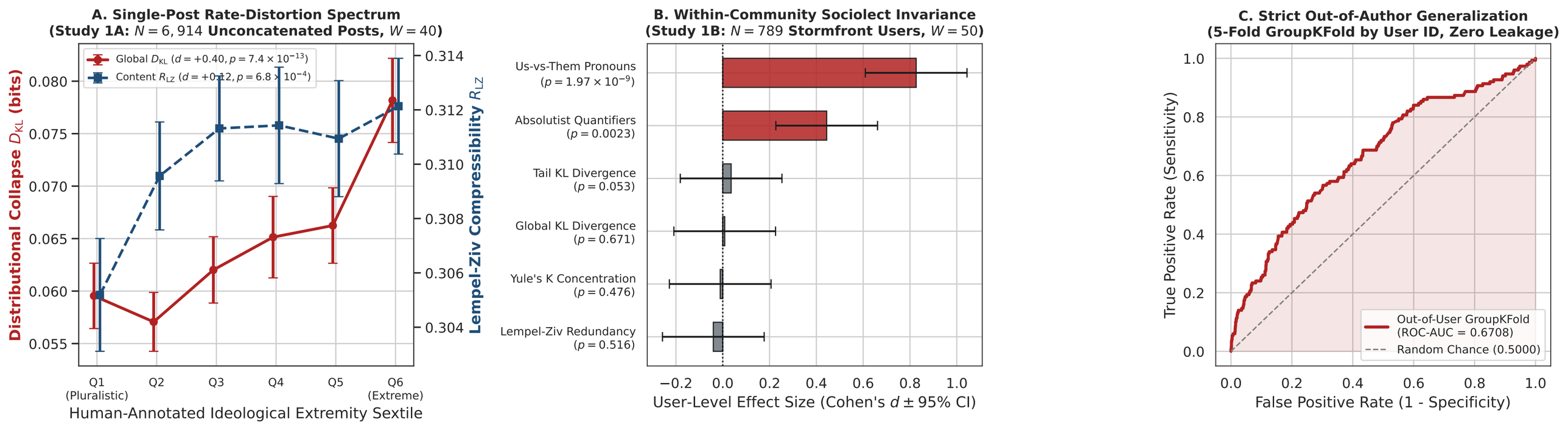
A. Single-Post Rate-Distortion Spectrum
(Study 1A: N = 6,914 Unconcatenated Posts, W = 40)
Global D_{KL} ($d = +0.40, p = 7.4 \times 10^{-13}$)
Content R_{LZ} ($d = +0.12, p = 6.8 \times 10^{-4}$)
Distributional Collapse D_{KL} (bits)
Lempel-Ziv Compressibility R_{LZ}
0.080
0.075
0.070
0.065
0.060
0.055
0.314
0.312
0.310
0.308
0.306
0.304
Q1 (Pluralistic)
Q2
Q3
Q4
Q5
Q6 (Extreme)
Human-Annotated Ideological Extremity Sextile
B. Within-Community Sociolect Invariance
(Study 1B: N = 789 Stormfront Users, W = 50)
Us-vs-Them Pronouns ($p = 1.97 \times 10^{-9}$)
Absolutist Quantifiers ($p = 0.0023$)
Tail KL Divergence ($p = 0.053$)
Global KL Divergence ($p = 0.671$)
Yule's K Concentration ($p = 0.476$)
Lempel-Ziv Redundancy ($p = 0.516$)
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
User-Level Effect Size (Cohen's $d \pm 95\%$ CI)
C. Strict Out-of-Author Generalization
(5-Fold GroupKFold by User ID, Zero Leakage)
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
0.0
0.2
0.4
0.6
0.8
1.0
Out-of-User GroupKFold (ROC-AUC = 0.6708)
Random Chance (0.5000)